

XRPL Validator — Daily Engineering Report • 2026-09-23

Generated 2026-09-24 00:21 EDT. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the live native-engine shadow, the reference rippled node, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **11 commits** (xrpl-validator 11)
- **16 engine findings fixed** (F383–F398)
- 3 test / vector commits
- Live shadow: 920/920 ledgers fully matched, 93,178 transactions, 0 engine receipts in the current soak
- Amendment **PermissionDelegationV1_1** lost majority at ledger 107,181,569 (2026-09-23 12:47 UTC)
- Amendment **BatchV1_1** held majority (since 2026-09-15 14:06 UTC); activation due ~2026-09-29 14:06 UTC

Live validator and native engine

Snapshot at generation time.

Measure	Value
Ledgers compared (current run)	920
Ledgers fully matched	920
Transactions applied natively	93,178
Result mismatches	0
Byte mismatches	0
Missing / extra keys	0 / 0
Batch inner-result mismatches	0
Engine receipts (current soak)	0
Live engine build running since	2026-09-23 23:02 EDT
Reference rippled	3.4.0, full, validated #107,196,061

Engine findings fixed

- **fix(ledger): F383-F388 — AMM deposit/withdraw sub-modes ported from rippled 3.4.0 (campaign 16)** (4789c7d , xrpl-validator) Testnet campaign 16 (AMM deposit/withdraw sub-modes against rippled 3.4.0) found six rules the engine lacked or got wrong: 16 of 138 bundles failed under BOTH engines (the transactors are shared). Each fix is a structural port — rippled's function body line for line, in its operation order and Number rounding modes, with its result codes in its order — on the signed Number port (tx::number) where the unsigned n_* ...
- **fix(ledger): F389-F394 — Batch inners: common preclaim, delegate permission, AccountTxnID order, AccountDelete fee, per-inner threading (campaign 23)** (74f5bbd , xrpl-validator) Campaign 23 (devnet, 2026-

09-23) crossed Batch (BatchV1_1 activates on mainnet ~09-29) with delegation, MPT, credentials, tickets, AccountDelete and LedgerStateFix inners. Six rules the engine had wrong: F389 (6-2, 6-5, 6-6, 6-7, 6-9) — a Batch inner runs rippled's whole per-transaction pipeline, common preclaim included (checkSeqProxy, checkPriorTxAndLastLedger — applySteps.cpp:179-183): an inner with a future/past ...

- **fix(ledger): F395-F396 — Batch touched sets read decoded pre-images; inner ids use the shared canonicaliser** (f7ad693 , xrpl-validator) A fresh-context review of F389-F394 against rippled 3.4.0 found two real problems (no critical one in the live-shadow path): F395 — tx/batch.rs built each inner's touched set with the pre-batch value from `state_map.lookup`, the RAW leaf. `state_replay` stores its leaves binary behind a `leaf_decoder`, so binary was compared with the sandbox's JSON: every unchanged write-back of a base object looked like a change (a ...
- **fix(ledger): F397 — NFTokenAcceptOffer preclaim in rippled's order: Destination before funds (soak #26 receipt)** (1e2d3d3 , xrpl-validator) Soak #26 receipt, mainnet #107190696 072B81D6348C: rKFauach accepted sell offer C4D78B6A, whose Destination is rpx9JThQ, while also short of the 7 XRP price. Mainnet: `tecNO_PERMISSION`. Ours: `tecINSUFFICIENT_FUNDS`. The bundle probe reproduced it on the live build.
- **fix(ledger): F398 — OfferStream original funds read through the afView's balance hooks (soak #27 receipt)** (ef17d8f , xrpl-validator) rippled's `strandFlow` opens `PaymentSandbox afView(&baseView)` beside the strand's own sandbox (`StrandFlow.h:133`), and `TOfferStreamBase` judges an unfunded or tiny offer "found" (permanently removed) only when `accountFundsHelper(cancelView_ = afView)` equals `ownerFunds_`. Both reads end in the `PaymentSandbox` hooks — `balanceHookIOU` (IOU lines and `xrpLiquid`'s full balance) and `ownerCountHook` — which walk the ...

Engineering changes

- **fix(node): UNL from unl.xrplf.org, retried until it loads; watch_engine counts canary lines apart** (74c3b50 , xrpl-validator) Since 2026-09-21 every live_viewer start came up with the consensus monitor frozen at `establish(round=1)`, UNL 0: the one startup fetch of the validator list failed on both sources, and the consensus loader answered "consensus disabled" for the life of the process. Both failures are TLS, both on the publishers' side: - vl.ripple.com, since its 2026-09-21 certificate renewal, serves the leaf (CN=*.ripple.com) twice ...

Tests and campaigns

- **test(node): campaign 23 — 60 byte-exact Batch × feature vectors; fetcher read-sets, build_batch.py** (e142bc1 , xrpl-validator)
- **test(node): probe_bundle judges a Batch whole — per-inner threading and inner verdicts** (a05e2dc , xrpl-validator)
- **test(node): campaign 24 — Batch × NFT, offer, check, channel, escrow, oracle, DID, AMM, trust line, AccountSet (27 vectors)** (ba89585 , xrpl-validator)

Docs and operations

- **docs: README — live-shadow status and progress history** (7857d61 , xrpl-validator)
- **docs: README — second clean 24-hour run; link the daily reports** (863ec8f , xrpl-validator)

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **PermissionDelegationV1_1 lost majority on mainnet.** The network applied `EnableAmendment` with `tfLostMajority` at ledger 107,181,569 (2026-09-23 12:47 UTC). It had held a majority since 2026-09-21 and would have activated around 2026-10-05; the two-week clock restarts only if it regains majority. **BatchV1_1 still holds its majority** (since 2026-09-15 14:06 UTC) and activates around 2026-09-29 14:06 UTC.
- **The second clean 24-hour run.** The live soak started 2026-09-22 11:38 EDT passed its 24-hour mark with zero engine receipts — every compared ledger matched rippled byte-for-byte (the first such run was 2026-09-20/21).
- **Batch conformance ahead of activation.** Campaign 23 (devnet) crossed Batch with delegation, MPT, credentials, tickets, AccountDelete and LedgerStateFix inners: six engine rules fixed (F389–F394). The subtlest: rippled threads an owner's AccountRoot (`threadOwners`) only for nodes an inner itself inserted or erased, in that inner's own ApplyStateTable — not for whatever the batch as a whole created. Campaign 24 (devnet) put NFT, offer, check, payment-channel, escrow, oracle, DID, AMM, trust-line and AccountSet inners inside Batches, each shaped so a later inner uses what an earlier one created: 27/27 byte-exact on the first run.
- **An independent review against the rippled 3.4.0 sources** of the Batch fixes found two more issues, fixed test-first (F395, F396): a replay-only pre-image read of raw leaves, and inner transaction ids hashed without the codec's canonical spelling of decimal MPT amounts and short-hex `AssetPrice`.
- **Desk checks for the two Batch-only bug classes found so far:** no other transactor copies raw transaction `Flags` into a ledger object (where `tfInnerBatchTxn` would leak), and none gates the fee level in preclaim (a Batch inner carries `Fee: "0"`).
- **Deployment.** The engine build f7ad693 (findings through F396) replaced the live shadow build at 15:31 EDT after its gate (1,589 suite tests and 706 flow-engine tests passing) and two mainnet replay windows (60 and 98 ledgers, zero mismatches). The post-deployment health check matched 402 ledgers with no mismatches, and a new 24-hour soak began at 15:57 EDT (seed ledger 107,187,825).
- **Receipt canary — first live run.** One drop was planted in the shadow's copy of a mainnet AccountRoot at ledger 107,188,304; the byte-for-byte compare flagged it and the receipt watcher logged it 255 seconds later. Canary lines are never counted as receipts: the check exists to prove that a zero-receipt soak means the comparison was running and sensitive. It now runs daily.
- **Soak #26 receipt and finding F397.** At ledger 107,190,696 an NFTTokenAcceptOffer was submitted by an account other than the offer's Destination, which was also short of the 7 XRP price. Mainnet returned `tecNO_PERMISSION`; the engine returned `tecINSUFFICIENT_FUNDS` — a fee-only failure either way. rippled judges the Destination before the funds; the engine's accept-offer preclaim now follows rippled 3.4.0's order exactly (F397, with the mainnet transaction pinned as a regression vector). Verified and merged; deployment awaits a decision.
- **Second deployment (F397).** Build 1e2d3d3 replaced the live shadow build at 20:51 EDT after its gate (1,594 suite tests and 706 flow-engine tests passing) and both mainnet replay windows (zero mismatches). The health check matched 404 ledgers with no mismatches, and soak #27 began at 21:17 EDT (seed ledger 107,192,792). Soak #26's four receipts — all the same NFTTokenAcceptOffer case, from one account retrying offers reserved for another — replay byte-exact on the fixed build and were archived.
- **Soak #27 receipt and finding F398.** At ledger 107,194,228 an immediate-or-cancel OfferCreate sold 942.58 XRP for RLUSD across one maker's five offers at a single quality. The payment engine's first pass consumed four of

them and left the fifth backed by about 10^{-14} RLUSD; on the second pass rippled found that offer too small to trade and removed it. rippled tells an offer *found* unfunded or tiny (removed) from one that *became* so during the payment (skipped) by comparing the owner's funds with the funds read through the strand's "all funds" view, whose balance reads apply the payment's deferred-credit table. The engine read that view's raw trust-line balance (1.5×10^{-14} against the adjusted 1.0×10^{-14}), judged the offer to have become tiny, and kept it. Both original-funds reads (trust line and XRP) now apply the same adjustments as rippled 3.4.0's `afView` (F398), with the mainnet transaction pinned as a regression vector. Verified and merged; deployment awaits a decision.

- **Third deployment (validator-list sources).** Build 74c3b50 replaced the live shadow build at 23:03 EDT after its gate (1,596 suite tests and 706 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches). The node now fetches its validator list from `unl.xrplf.org` as well as `vl.ripple.com`, verifies the publisher's signature, and retries with backoff when no list can be fetched instead of running without consensus tracking. The list loaded from `unl.xrplf.org` (35 validators) and the consensus monitor resumed. The health check matched 411 ledgers with no mismatches, and soak #28 began at 23:29 EDT (seed ledger 107,194,842). Soak #27 ended at the deployment with the single receipt above.
- **Deployment practice.** The node's start-up state sync (19.9 million ledger objects, read from the reference rippled node) took 19.4 minutes instead of the usual 5.5–7, because a test build was running on the reference node's machine at the same time. Builds and replays there are now held automatically until a deployment's health check completes.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- `PermissionDelegationV1_1` lost majority — ledger 107,181,569, 2026-09-23 12:47 UTC

Amendment	Majority since	Activation due
BatchV1_1	2026-09-15 14:06 UTC	2026-09-29 14:06 UTC

Amendments enabled on mainnet: 94.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Hub-Mediated Multilateral Check Netting** (settlement) Add transaction `ttCHECK_NET` carrying `sfCheckIDs` (`STVector256`, bounded ≤ 8) referencing existing Check objects where the submitting account is either `sf`
- **On-Ledger Settlement Finality Receipts** (settlement) Add an optional `tfFinalityReceipt` flag to `EscrowFinish` and `CheckCash`; when set and the discharge succeeds, `doApply` additionally creates a compact `ltSE`

- **Destination-Enforced Travel-Rule Envelope** (compliance) Add `lsfRequireTravelRule` (AccountSet-settable flag) and `sfTravelRuleThreshold` to AccountRoot, plus a new bounded VL SField `sfTravelRuleEnvelope` (max 4

Appendix — commits

Time	Repo	Commit	Subject
01:54	xrpl-validator	4789c7d	fix(ledger): F383-F388 — AMM deposit/withdraw sub-modes ported from rippled 3.4.0 (campaign 16)
02:57	xrpl-validator	e142bc1	test(node): campaign 23 — 60 byte-exact Batch × feature vectors; fetcher read-sets, build_batch.py
02:57	xrpl-validator	74f5bbd	fix(ledger): F389-F394 — Batch inners: common preclaim, delegate permission, AccountTxnID order, AccountDelete fee, per-inner threading (campaign 23)
03:15	xrpl-validator	a05e2dc	test(node): probe_bundle judges a Batch whole — per-inner threading and inner verdicts
03:47	xrpl-validator	ba89585	test(node): campaign 24 — Batch × NFT, offer, check, channel, escrow, oracle, DID, AMM, trust line, AccountSet (27 vectors)
04:21	xrpl-validator	f7ad693	fix(ledger): F395-F396 — Batch touched sets read decoded pre-images; inner ids use the shared canonicaliser
10:42	xrpl-validator	7857d61	docs: README — live-shadow status and progress history
12:25	xrpl-validator	863ec8f	docs: README — second clean 24-hour run; link the daily reports
19:05	xrpl-validator	1e2d3d3	fix(ledger): F397 — NFTTokenAcceptOffer preclaim in rippled's order: Destination before funds (soak #26 receipt)
22:08	xrpl-validator	74c3b50	fix(node): UNL from unl.xrplf.org, retried until it loads; watch_engine counts canary lines apart
23:30	xrpl-validator	ef17d8f	fix(ledger): F398 — OfferStream original funds read through the afView's balance hooks (soak #27 receipt)